# LOVELY PROFESSIONAL UNIVERSITY

## Sample Question Set - Unit 4

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO4
Paper: Sample Multiple Choice Questions - ANSWER KEY
Maximum Marks: 40 x 1 = 40

Unit: 4
Unit Title: Evolution of Modern AI and Foundation Models
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

**Answer Grid (Key)**

| Q. No. | Answer | Q. No. | Answer | Q. No. | Answer | Q. No. | Answer |
|---|---|---|---|---|---|---|---|
| Q1 | A | Q11 | D | Q21 | A | Q31 | C |
| Q2 | B | Q12 | B | Q22 | C | Q32 | A |
| Q3 | C | Q13 | D | Q23 | B | Q33 | A |
| Q4 | D | Q14 | C | Q24 | D | Q34 | D |
| Q5 | B | Q15 | B | Q25 | C | Q35 | B |
| Q6 | D | Q16 | A | Q26 | D | Q36 | C |
| Q7 | A | Q17 | B | Q27 | A | Q37 | C |
| Q8 | C | Q18 | A | Q28 | B | Q38 | B |
| Q9 | C | Q19 | D | Q29 | D | Q39 | D |
| Q10 | A | Q20 | C | Q30 | B | Q40 | A |

### Multiple Choice Questions (40 x 1 = 40 Marks)

**Q1. Name the deep learning architecture designed for image data.**

A) The convolutional neural network <-- correct
B) The recurrent neural network
C) The decision tree ensemble
D) The production rule system

**Answer: A.** Convolutional layers detect local patterns wherever they occur in an image, which is what makes them suited to vision tasks.

**Q2. Identify the type of machine learning in which the system learns from a reward received after each action.**

A) Supervised learning
B) Reinforcement learning <-- correct
C) Unsupervised learning
D) Semi-supervised learning

**Answer: B.** The reward says how well the agent did, not what it should have done, so the agent has to discover a good policy by trying actions.

**Q3. State what Edge Artificial Intelligence means.**

A) Running the model on a remote cluster of servers
B) Training the model on data drawn from many devices
C) Running the model on the device that collects the data <-- correct
D) Running the model only at the edge of a network cable

**Answer: C.** Inference happens locally, which removes the round trip to a server and keeps the raw data on the device.

**Q4. Name the technique that reduces the numeric precision used to store a model's weights.**

A) Pruning
B) Fine-tuning
C) Augmentation
D) Quantisation <-- correct

**Answer: D.** Quantisation stores each weight in fewer bytes, cutting memory and often speeding inference at the cost of a little accuracy.

**Q5. Identify what a foundation model is pre-trained on.**

A) A small dataset from one specialised domain
B) A very large and broad body of general data <-- correct
C) A labelled dataset for one narrow task
D) The outputs of an earlier expert system

**Answer: B.** Broad pre-training is what gives the model general capability that can then be adapted cheaply to many particular tasks.

**Q6. State the three types of machine learning named on the syllabus.**

A) Symbolic, statistical and hybrid learning

B) Shallow, deep and very deep learning

C) Local, edge and cloud based learning

D) Supervised, unsupervised and reinforcement learning <-- correct

**Answer: D.** The three are distinguished by the training signal each receives: a correct answer, no answer at all, or a reward following an action.

**Q7. Explain why symbolic Artificial Intelligence was described as brittle.**

A) It produced no answer for a case its rules did not cover <-- correct

B) It ran too slowly to be used on real problems

C) It required an extremely large training dataset

D) It could not represent any form of knowledge

**Answer: A.** A rule base works well inside the situations its authors anticipated and fails outright just outside them, with no graceful degradation.

**Q8. Distinguish training from inference in the life of a model.**

A) Inference fits the parameters, training applies them

B) Both of the two fit the parameters of the model

C) Training fits the parameters, inference applies them <-- correct

D) Both of the two apply the parameters to new data

**Answer: C.** Training is done once and is expensive; inference is done repeatedly in service, and it is inference cost that dominates a deployed system.

**Q9. Describe what an agentic Artificial Intelligence system does beyond answering a question.**

A) It answers the same question much more quickly

B) It stores every question it has ever been asked

C) It plans steps, calls tools and revises its own plan <-- correct

D) It refuses any question outside its training data

**Answer: C.** An agentic system decomposes a request, acts through external tools, inspects the results and adjusts, rather than producing one reply and stopping.

**Q10. Interpret the following as a type of machine learning: a system is given customer records each carrying a known outcome of whether that customer left.**

A) Supervised learning <-- correct

B) Unsupervised learning

C) Reinforcement learning

D) Semi-supervised learning

**Answer: A.** Every training example carries the correct answer, so the task is to learn the mapping from the features to that known outcome.

**Q11. Predict the effect on latency and on privacy of running a model on the device rather than in the cloud.**

A) Latency rises and the raw data must leave the device

B) Latency falls but the raw data must leave the device

C) Latency rises but the raw data stays on the device

D) Latency falls and the raw data need not leave the device <-- correct

**Answer: D.** Removing the network round trip is the latency gain, and keeping the data local is the privacy gain. The cost is a smaller model on limited hardware.

**Q12. Recognize the deep learning application described: a system converts spoken audio into written text.**

A) Image classification

B) Speech recognition <-- correct

C) Object detection

D) Recommendation

**Answer: B.** Speech recognition is one of the standard applications of deep learning, alongside vision, language and recommendation.

**Q13. Explain why deep learning became practical only comparatively recently.**

A) The underlying mathematics was only just discovered

B) Symbolic methods were legally prohibited

C) Neural networks were invented in the last decade

D) Large datasets and fast parallel hardware became available <-- correct

**Answer: D.** The ideas are decades old. What changed was the volume of digital data and the arrival of graphics processors able to train large networks in reasonable time.

**Q14. Distinguish a foundation model from a model trained for one particular task.**

A) A task model is broad and then adapted to one task

B) Both of the two are trained for one task only

C) A foundation model is broad and then adapted to tasks <-- correct

D) Both of the two are trained on general data only

**Answer: C.** The foundation approach separates an expensive general pre-training, done once, from a cheap adaptation repeated for each downstream use.

**Q15. Explain what it means for an Artificial Intelligence model to be multimodal.**

A) It can be deployed on more than one kind of device

B) It accepts more than one kind of input, such as text and images <-- correct

C) It was trained by more than one organisation

D) It produces more than one answer per question

**Answer: B.** A multimodal model relates the content of an image to the words of a question about it, which a single-modality pipeline can only do awkwardly.

**Q16. Describe the constraint that distinguishes Tiny Artificial Intelligence from Edge Artificial Intelligence.**

A) Tiny AI targets microcontrollers with very little memory and power <-- correct

B) Tiny AI targets servers with very large memory

C) Tiny AI is trained on a much smaller dataset

D) Tiny AI produces much shorter answers

**Answer: A.** Both run inference locally, but Tiny AI works within kilobytes of memory and milliwatts of power on a sensor or a microcontroller.

**Q17. Predict what a statistical model does when it meets an input quite unlike anything in its training data.**

A) It reports that the input is outside its experience

B) It still produces an output, with no warning of the difference <-- correct

C) It refuses to produce any output whatsoever

D) It automatically retrains itself on the new input

**Answer: B.** The model has no notion of the boundary of its own competence, which is why monitoring for drift and out-of-distribution input is a deployment requirement.

**Q18. Summarize what statistical Artificial Intelligence supplied that the symbolic approach could not.**

A) Rules inferred from examples rather than stated by hand <-- correct

B) Guaranteed correctness for every possible case

C) An explanation of every decision it produces

D) Freedom from any need for training data

**Answer: A.** Learning from data made it possible to build systems for tasks nobody can articulate as rules, at the cost of explainability and of any guarantee.

**Q19. Explain why a recurrent architecture suits data that arrives as a sequence.**

A) It processes every position independently

B) It requires the sequence to be of fixed length

C) It converts the sequence into a single image

D) It carries information forward from earlier positions <-- correct

**Answer: D.** The hidden state passed from step to step lets the network condition its output on what came before, which is what sequential data requires.

**Q20. Distinguish pruning from quantisation as ways of making a model smaller.**

A) Quantisation removes weights, pruning stores them in fewer bits

B) Both of the two remove weights from the network

C) Pruning removes weights, quantisation stores them in fewer bits <-- correct

D) Both of the two reduce the precision of the weights

**Answer: C.** The two are complementary and are frequently applied together when a model must be fitted onto constrained hardware.

**Q21. Solve the following: a model holding 3 billion parameters is stored at 2 bytes for each parameter. Determine the memory its weights occupy.**

A) 6 GB <-- correct

B) 3 GB

C) 1.5 GB

D) 12 GB

**Answer: A.** 3 billion x 2 bytes = 6 billion bytes, which is approximately 6 GB.

**Q22. Compute the memory saved by quantising a model from 2 bytes per parameter to 1 byte.**

A) A quarter of the original

B) Three quarters of the original

C) Half of the original <-- correct

D) None of the original

**Answer: C.** Halving the bytes per parameter halves the storage. The accuracy usually falls only slightly, which is what makes the exchange worthwhile.

**Q23. Apply the three types of machine learning to a task in which fifty thousand records carry no labels and the request is to divide them into groups.**

A) Supervised learning

B) Unsupervised learning <-- correct

C) Reinforcement learning

D) Transfer learning

**Answer: B.** No target exists, so nothing can be predicted; the structure has to be discovered from the features alone.

**Q24. Use the distinction between pre-training and fine-tuning to determine which of the two is repeated for each new domain.**

A) Pre-training

B) Both of the two equally

C) Neither of the two

D) Fine-tuning <-- correct

**Answer: D.** Pre-training is done once on broad data at very high cost. Fine-tuning is the cheap step repeated whenever the model must be adapted to a new domain.

**Q25. Examine the following and determine the cause: a vision model trained on daylight photographs performs badly on images taken at night.**

A) The model has too many parameters to work

B) The images were stored in the wrong file format

C) The training data did not cover that condition <-- correct

D) The model was quantised before deployment

**Answer: C.** A statistical model reproduces the conditions of its training set. Coverage of the

deployment conditions is a data problem, not a modelling one.

**Q26. Solve the following: a hosted model charges 0.0015 dollars for each query and 40,000 queries are made in a month. Determine the monthly cost.**

A) 6 dollars
B) 600 dollars
C) 40 dollars
D) 60 dollars <-- correct
**Answer: D.** 0.0015 x 40,000 = 60 dollars, which is the figure to compare against the cost of hosting an open model locally.

**Q27. Apply Edge Artificial Intelligence to a setting in which the network connection is frequently unavailable, and determine the benefit.**

A) The system keeps working while the connection is down <-- correct
B) The system trains itself faster while offline
C) The system needs a much larger model to run
D) The system stops needing any sensors at all
**Answer: A.** Local inference removes the dependence on connectivity, which is decisive for vehicles, remote installations and safety functions.

**Q28. Compare a convolutional network with a recurrent network on the kind of data each is suited to.**

A) Recurrent suits images, convolutional suits sequences
B) Convolutional suits images, recurrent suits sequences <-- correct
C) Both of the two are designed for image data
D) Both of the two are designed for sequential data
**Answer: B.** Convolutions exploit spatial locality; recurrence exploits order in time. Matching the architecture to the structure of the data is the first design decision.

**Q29. Use transfer learning on a task holding only 500 labelled images and determine the usual procedure.**

A) Train every layer of the network from the beginning
B) Discard the pre-trained network and use rules instead
C) Collect no further data and accept the poor result
D) Keep the pre-trained feature layers and retrain the last ones <-- correct
**Answer: D.** The early layers hold general visual features that transfer across tasks. Only the task-specific head needs the small labelled set.

**Q30. Compute the memory occupied by 1.5 billion parameters stored at 4 bytes each.**

A) 1.5 GB
B) 6 GB <-- correct
C) 3 GB
D) 12 GB
**Answer: B.** 1.5 billion x 4 bytes = 6 billion bytes, approximately 6 GB, the same total as 3 billion parameters at 2 bytes.

**Q31. Examine the following and determine the remedy: a general purpose model answers everyday questions well but handles the wording of medical reports poorly.**

A) Increase the number of queries it is asked
B) Reduce the precision of its stored weights
C) Adapt it with domain data or retrieve from domain documents <-- correct
D) Deploy it on a device rather than a server
**Answer: C.** Its training corpus under-represents that specialist language. Fine-tuning on domain text, or grounding it in retrieved domain documents, addresses the gap.

**Q32. Apply the definition of agentic Artificial Intelligence to determine which system qualifies.**

A) One that plans steps, uses tools and revises its plan <-- correct
B) One that answers a single question and stops
C) One that stores its answers in a database
D) One that runs entirely on a small device
**Answer: A.** The defining features are decomposition of a task, action through external tools, and revision of the plan in the light of what those tools return.

**Q33. Use the three types of machine learning to classify a system given labelled photographs of defective and sound welds.**

A) Supervised learning <-- correct
B) Unsupervised learning
C) Reinforcement learning
D) Agentic learning
**Answer: A.** Each example carries the correct class, so the system learns the mapping from image to label. This is a supervised classification task.

**Q34. Solve the following: a model stored at 4 bytes per parameter is quantised to 1 byte per parameter. Determine the resulting size as a percentage of the original.**

A) 50 per cent
B) 75 per cent
C) 12.5 per cent
D) 25 per cent <-- correct
**Answer: D.** One byte in place of four leaves a quarter of the original storage, so the model occupies 25 per cent of the memory it did before.

**Q35. Compare Edge Artificial Intelligence with cloud inference on the network bandwidth each one consumes.**

A) The cloud sends little or nothing, edge sends every input

B) Edge sends little or nothing, the cloud sends every input <-- correct

C) Both of the two send every input over the network

D) Neither of the two uses any network bandwidth

**Answer: B.** A camera running locally may transmit only an occasional alert, whereas cloud inference must upload every frame it wishes to have analysed.

**Q36. Examine the risk of fine-tuning a general model on a very small specialised dataset.**

A) It becomes unable to run on any device

B) It loses the ability to accept text input

C) It may lose general ability while overfitting the small set <-- correct

D) It requires the pre-training to be repeated

**Answer: C.** This is catastrophic forgetting. Guarding against it means keeping some general data in the mix, limiting the training, or using retrieval instead.

**Q37. Apply the idea of distillation and determine what it produces.**

A) A larger model built from several small ones

B) A model with its weights stored in fewer bits

C) A smaller model trained to imitate a larger one <-- correct

D) A model with its least useful weights removed

**Answer: C.** The small student model is trained on the outputs of the large teacher, which often transfers much of the capability into far fewer parameters.

**Q38. Use the relation between model size and serving cost to determine the effect of halving the number of parameters.**

A) The memory needed at inference is roughly doubled

B) The memory needed at inference is roughly halved <-- correct

C) The memory needed at inference is unchanged

D) The memory needed at inference falls to zero

**Answer: B.** Weight storage is the dominant term at inference, so it scales with the parameter count. This is why size reduction is the first lever in deployment.

**Q39. Demonstrate how a multimodal model is applied to a task that combines a photograph with a question written about it.**

A) It answers from the text and ignores the photograph

B) It answers from the photograph and ignores the text

C) It answers the two inputs as two separate questions

D) It represents both inputs together and answers from the pair <-- correct

**Answer: D.** Relating the two modalities is exactly what a single multimodal model provides, and what a pipeline of separate models can only approximate.

**Q40. Relate the volume of training data available to the choice between a deep network and a classical method.**

A) Scarce data favours a classical method or transfer learning <-- correct

B) Scarce data always favours a very deep network

C) The volume of data has no bearing on the choice

D) Abundant data always favours a classical method

**Answer: A.** Deep networks have many parameters and need many examples. With a few hundred records a simpler model, or a pre-trained network adapted, is the sound choice.

**----- End of Sample Question Set -----**